

Choosing glaucoma surgery and reading its evidence

Part 2 | A surgical target must match the disease burden

The lowest-risk intervention is not necessarily the one that meets the target.

Planning factor	Why it matters
Progression rate and remaining field	Defines the cost of inadequate lowering
Target pressure and baseline IOP	Constrains likely treatment success
Conjunctiva and prior operations	Changes surgical options and prognosis
Medication burden and follow-up capacity	Changes feasibility and preferences

Discuss expected efficacy, complications and the postoperative workload. NICE recommendations apply to their specified populations and health-system context; they are not a claim that every patient must receive the same procedure. Revisit a target when progression persists despite apparently acceptable clinic measurements.

TVT applies to an operated-eye population

Eligibility is part of the result, not a footnote.

TVT feature	Inference boundary
Prior cataract or filtering surgery	Do not treat as wholly surgery-naive evidence
Tube versus trabeculectomy with MMC	Compare the actual procedures studied
Defined failure composite	Inspect which events count as failure
Multi-year follow-up	Separate early from durable outcomes

A randomized trial can have strong internal validity while not answering every surgical setting. Failure criteria may combine pressure, reoperation and loss-of-light-perception outcomes. Look at medication use and complications alongside the headline success rate; a lower failure rate does not imply superiority on every endpoint.

PTVT asks a different population question

Do not merge trial names into one universal tube-versus-trab rule.

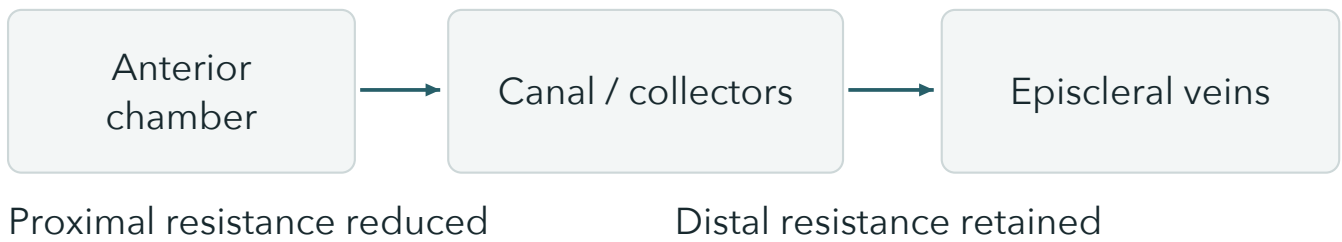
PTVT feature	Decision relevance
No previous incisional ocular surgery	Different starting population from TVT
Randomized procedure comparison	Balances measured and unmeasured baseline factors
Pressure and medication outcomes	May not move in the same direction
Complications and reoperations	Part of shared decision-making

Compare eligibility, definitions, follow-up and effect estimates before applying results. The right trial for a particular eye may be the one whose population and operative context resemble it, not the one with the most memorable headline. A treatment effect is not automatically transferable across scarred and untouched conjunctiva.

Trabecular surgery retains downstream constraints

Opening the proximal route does not bypass every resistance.

Trabecular route



Bleb route



Original map: trabecular procedures retain distal canal and venous pathways; a bleb bypasses them.

Trabecular MIGS can reduce resistance and medication burden in suitable eyes, but distal outflow and episcleral venous pressure remain relevant. A very low target in advanced disease may require a different route. The word microinvasive describes an approach, not a guaranteed pressure floor or equal evidence for every device.

Use endpoint-specific counseling

A single success percentage hides the tradeoffs patients need.

Endpoint	Patient-facing question
Pressure control	Does it plausibly protect the remaining field?
Drop reduction	Is burden reduced even if pressure is similar?
Reoperation	What happens if the first procedure is insufficient?
Safety and recovery	What risks and follow-up are acceptable?

Avoid combining unlike definitions of success across studies. Discuss the likelihood that additional treatment will be needed and how that would be chosen. A comfortable postoperative eye can still have insufficient pressure control; an excellent pressure result can still carry surface, visual or infection burdens.